

Diagnosing Accuracy Limitations in Constraint-Based IP Geolocation

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Abstract

The “CBG++” algorithm for active geolocation estimates the geographic location of a *target* host to be within the intersection of several disks drawn on the globe, centered on *landmarks* in known locations, with radii derived from packet round-trip timing. Using 788 RIPE Atlas anchors as both landmarks and targets, 98% of all predicted regions overlap with the country where the target host is located. However, only 77% of predicted regions include the true location of the target, and the centroid of the predicted region is within 40 km of the true location—roughly city scale—for only 65% of the regions.

We experimented with two narrow adjustments to CBG++ to try to improve its city-scale accuracy. Neither is effective. CBG++’s linear estimate of the delay-distance relationship (its “calibration”) is conservatively accurate: 99.72% of individual disks contain the true location of the target, but many of them overshoot it. Reducing overshoot by tightening the calibration often causes the disks not to overlap at all. When this happens, CBG++ falls back to an even more conservative estimate (bounded only by physical limits) so the effect is to make the predicted regions 2–7 $\times$ larger, and move their centroids *farther* from the true locations.

Leaving the calibration alone, but removing disks that do not contain the true location—even with post-hoc knowledge of which ones those are—*also* enlarges the predicted regions without improving city-scale accuracy. This is because disks that miss the target are usually only short by a few tens of kilometers. Treating them as completely wrong discards useful constraints.

1 Introduction

IP geolocation identifies the physical location of an Internet host, usually referred to as the *target*. It is useful for a wide range of applications, from the technical (network diagnostics) to the economic (targeted advertising, services legal in only some locations) [2, 4, 5, 7]. There are two broad classes of technique for IP geolocation. *Passive* geolocation relies on databases published by various commercial services, and/or self-reports by host operators [3, 16]. It is efficient and precise, but inaccurate when a target’s operators are actively trying to conceal its location [20]. *Active* geolocation instead infers the target’s location from packet round-trip time (RTT) measurements between the target and *landmark* hosts in known locations. It is slower and less precise, but harder to spoof.

Active geolocation is easy when there is a landmark close to the target. Several recent systems exploit this observation by prioritizing nearby, low-RTT landmarks, and by finding such landmarks efficiently [6, 8, 9, 12, 13, 18]. However, RIPE IPmap [8] achieved only 75.8% coverage over 16,245 reachable interconnection addresses using its default 10 ms RTT threshold. Tightening that threshold

improves accuracy but sharply reduces how many targets can be located: coverage falls to 51.1% at 5 ms and 24.1% at 2 ms.

When no single landmark is close enough to be relied on by itself, one must combine information from several landmarks. Constraint-Based Geolocation (CBG) [11] is one of the oldest published algorithms for this. It uses linear estimates of the relationship between RTT and distance traveled to draw disks on the surface of the earth, one for each landmark; then it takes the intersection of all these disks as its prediction of the target’s location. CBG fails if the disks don’t all intersect. An incremental refinement, CBG++ [20], avoids failure by favoring distance bounds that are less likely to underestimate how far a packet could have traveled, and by identifying the largest group of disks whose intersection is not empty.

In a cross-validation test of 788 RIPE Atlas anchors, CBG++’s predicted regions overlap the target’s true country in 98% of cases, but they only include the target’s exact location in 77% of cases, and the region’s centroid is within 40 km of the target in only 65% of cases. Thus, CBG++ cannot provide accurate city-scale predictions for about a third of targets. In this paper, we investigate why not, by experimenting with two narrow adjustments to the algorithm. Neither adjustment improves the results, but their failure clarifies what the problem is and what might work better.

First, we observe that CBG++ is designed to overestimate the distance each packet could have traveled, in order to reduce the odds of an empty intersection. A total of 99.72% of the landmark disks it generates contain the true target location, but they are loose: the median disk’s radius is 2.1 $\times$ the true distance to the target, and 70% are discarded outright as too large to be useful. Several active geolocation algorithms published since CBG [e.g. 15, 21] propose tighter, more sophisticated models of the RTT–distance relationship. In Section 3, we test a tighter, but still linear, RTT–distance model based on quantile regression, and find that this makes the final localization *worse*. Mutual overlap between CBG++’s “bestline” disks is reduced and it falls back to “baseline” disks, producing final regions 2–7 $\times$ larger.

Second, we note that only 77% of the final intersected regions contain the target’s true location. This must somehow be due to the 0.28% of disks that miss the true location. In Section 5, we experiment with removing those disks, using two realizable algorithms and an oracle. This also tends to make the final localization worse. Disks that don’t contain the true location, we find, are likely to be only *slightly* too small, and thus still provide useful constraints on what is *near* the true location.

Together, these results suggest that true improvements will come from modeling uncertainty in individual landmarks’ distance estimates, rather than treating each estimate as a hard limit and trying to filter out “bad” estimates.

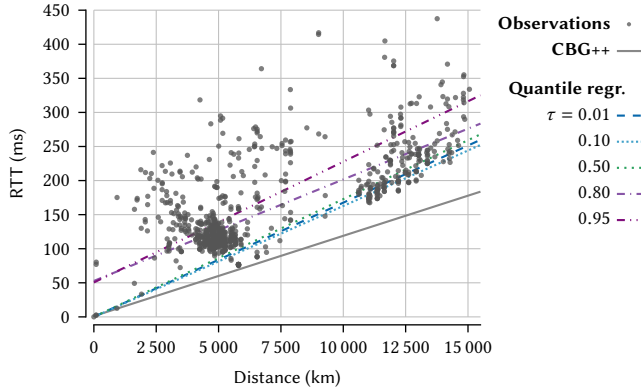


Figure 1: CBG++ bestline and QR fits for five values of τ . Anchor 6798 (Dubai), 2026 Jan 22.

2 Background: CBG++

Constraint-Based Geolocation (CBG) [11] estimates the RTT–distance relationship separately for packets sent from each landmark, using a linear model based on RTT measurements between that landmark and all the others. The model is fit via linear programming, selecting the steepest line between two data points that does not pass above any other data point. An example CBG calibration data set and linear model are shown in Figure 1 (along with several potential tighter estimates, discussed in Section 4).

CBG’s model can easily underestimate the distance a packet could have traveled, for instance if there was heavy network load during calibration. Distance underestimates produce disks that don’t intersect any other disks, causing geolocation to fail.

CBG++ [20] seeks to eliminate failures due to underestimation using two algorithmic refinements. During calibration, CBG++ imposes plausibility constraints: each modeled line (the *bestline* for that landmark) must have a slope between 84.5 and 200 km/ms. These reflect physical limits: 200 km/ms is the speed of light in optical fiber, and 84.5 km/ms is the apparent speed of a packet that was relayed by a geostationary satellite, regardless of how far apart the surface stations were. During geolocation, CBG++ computes two disks for each landmark, a small one using that landmark’s bestline, and a larger one using a fixed 200 km/ms *baseline*. Disks covering more than half the Earth’s surface are discarded as too large to be useful. The algorithm then finds the largest subset of the remaining *baseline* disks with a non-empty intersection. *Bestline* disks that do not overlap with that baseline localization are discarded as too small, and replaced with the corresponding *baseline* disk. Finally, the algorithm finds the largest subset of the remaining disks with a non-empty intersection, and uses that intersection as its prediction.

3 Evaluating CBG++ on RIPE Anchors

We evaluated CBG++ by cross-validation on 788 RIPE Atlas anchors: all that were operational at the time of the study. Anchors have public, verified locations and continuously collect anchor-to-anchor RTT measurements, which are made available in a public database [19]. We use each anchor as a target in turn, and compare CBG++’s prediction to its known location.

Table 1: Overall performance of CBG++ on 788 RIPE Atlas anchors (measured Jan 22, 2026).

Category	Metric	Value
Accuracy	Region overlaps true country	98.1 %
	Region contains true location	77.0 %
	Centroid ≤ 40 km from true location	65.5 %
Precision	Median region area	3 261 km ²
	Median centroid distance	12.6 km
	Median edge distance	25.6 km
	Median countries overlapped	1

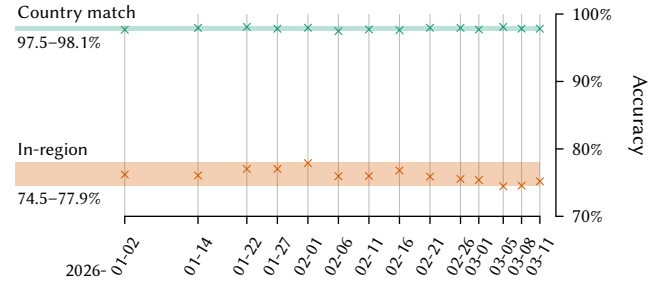


Figure 2: Accuracy of CBG++ over 10 weeks (Jan–Mar 2026).

There are many ways to assess the quality of a prediction region. Table 1 shows CBG++’s score on seven possible metrics, categorized as measuring accuracy or precision. This paper focuses on the accuracy metrics: whether the region overlaps the true country (*country matching*), whether it includes the true location of the target (*in-region containment*), and whether the region’s centroid is close enough to the true location to identify a specific locality (*city matching*). We also discuss the predicted region’s size and the distance from the centroid to the true location in several places; as is common, improvements to accuracy may harm precision.

At one moment in early 2026, when tested on the 788 active RIPE Atlas anchors, CBG++ achieved 98.1% country matches, but only 77.0% in-region containment. To determine whether this is a stable result, we repeated the cross-validation test 14 times, spread over 10 weeks, using the same anchors. Figure 2 shows country matching and in-region containment for all 14 dates. In-region containment shows some variation but remains in a narrow band of 74.5–77.9%, while country-level matching is stable at $\sim 98\%$.

4 Tighter Calibration Does Not Help

Precise prediction regions are produced by disks *just* large enough to include the target’s true location. However, in the original CBG, if any disks were too small, geolocation would *fail*, as there would be no region overlapped by all the disks. CBG++ therefore errs on the side of allowing disks to be too large.

Prior work improves on CBG by refining the model of the RTT–distance relationship to produce tighter constraints on the target’s location [15, 21], but makes other changes at the same time. To isolate the effect of tighter constraints, we changed CBG++’s calibration to use linear quantile regression (QR), leaving everything

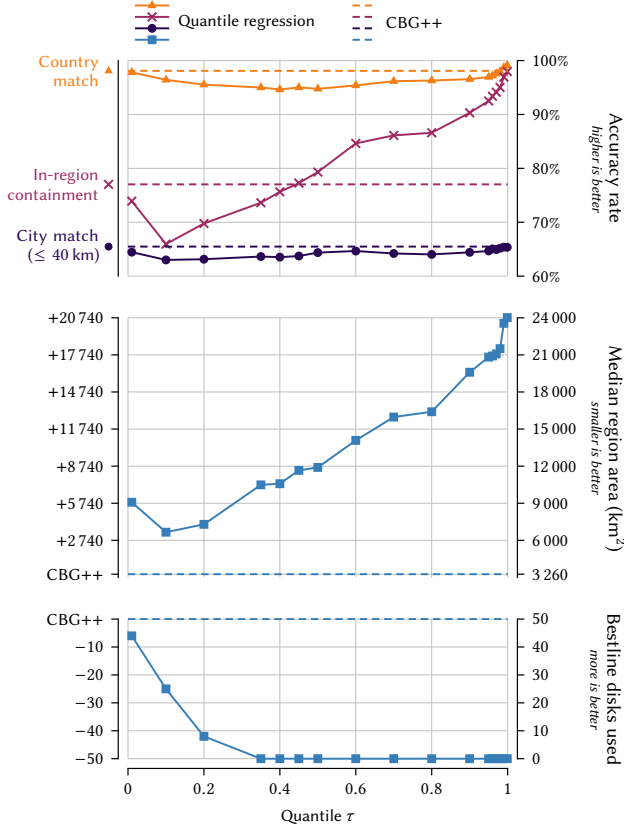


Figure 3: CBG++ vs. quantile-regression (QR). Top panel: In-region containment rises with τ , while city-level accuracy remains static. Middle: QR-based prediction regions are inflated 2–7 $\times$ relative to unmodified CBG++. Bottom: Both effects are because tighter bestline disks lose mutual overlap, causing intersection to fall back to baseline disks.

else the same. QR estimates the τ 'th quantile of a dependent variable; larger values of τ permit more of the calibration data points to fall below the bestline. The dashed lines in Figure 1 illustrate the effect of several values of τ on one landmark's calibration.

We tested a wide range of values for τ against the same set of anchors as before. Figure 3 shows the effects. The top panel shows accuracy metrics. Larger values of τ steadily improve in-region containment, after an initial dip, to a maximum of 98% at $\tau = 0.999$. But the effect of QR on country and city matching is weak, *negative*, and uncorrelated with τ . The middle panel shows the cost of improved in-region containment: the median size of the predicted region rises steadily with larger τ , to a maximum of 24 000 km², 6.6 $\times$ larger than the median CBG++ region area.

Finally, the bottom panel looks “under the hood” and reveals that as τ rises, CBG++’s intersection process finds more and more of the *bestline* disks to be too small to overlap each other, and replaces them with larger *baseline* disks. Our attempt to tighten the calibration backfired, making it looser. We infer from this that tighter calibration must be coupled with a more sophisticated intersection process, perhaps (following Spotter [15]) one that explicitly allows for uncertainty in how big each disk should be.

5 Excluding “Bad” Disks Does Not Help

CBG++’s “largest subset with a nonempty intersection” rule ensures it will always deliver a prediction, but it is still sensitive to disks that are too small. If each of the *bestline* disks that CBG++ includes in an intersection is considered by itself, 99.72% of them contain the true location of the target. Yet only 77% of the final regions do, because just one disk that excludes the true location of the target—perhaps missing it by only a few kilometers—is enough to cause an in-region containment failure.

Table 2 shows that, of 185 such failures from the Jan 22 evaluation run, only one disk was at fault in 168 of the cases (90.8%). We experimented with trying to identify and remove these disks. Prior work emphasizes the value of landmarks very close to the target [6, 13], so we tried two algorithms that assume these are more likely to be the right size. *Trusted-core then expand* requires the K shortest-RTT disks to be included, *RTT-weighted maximum subset* assigns each disk influence inversely proportional to its size. (Details of these algorithms have been omitted for space reasons.) Neither escapes the usual trade-off. *RTT-weighted maximum subset* raises region containment from 77.0% to 81.0%, but enlarges the median region from 3 261 to 4 886 km²; *trusted-core then expand* shrinks the median region to 2 492 km², but containment falls to 70.8%. City-level accuracy moves by less than 0.3% either way.

These algorithms do not work because the assumption is wrong: it is the *smallest* disks that are most likely to be too small. As shown in Figure 4, 31% of disks that miss the true location are within 50 km of it, and 64% are within 500 km. 3.1% of the disks generated by landmarks within 50 km of the target are too small, compared with

Table 2: Number of selected disks that exclude the target, for each in-region containment failure (Jan 22, 2026).

Disks	Targets affected
1	168 90.8%
2	10 5.4%
3	3 1.6%
4–8	4 2.2%

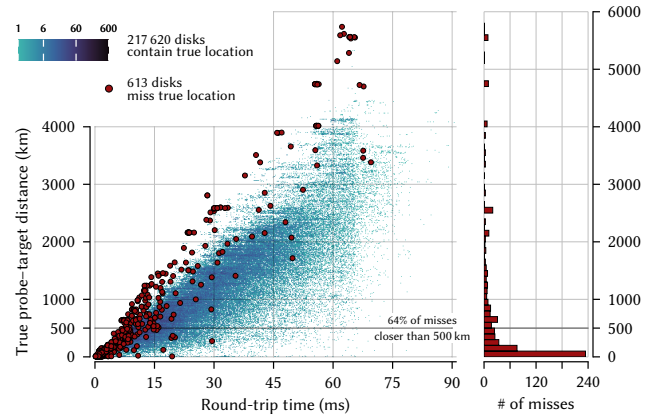


Figure 4: Per-disk containment rate for CBG++ empirical disks (LP bestline) on Jan 22, 2026.

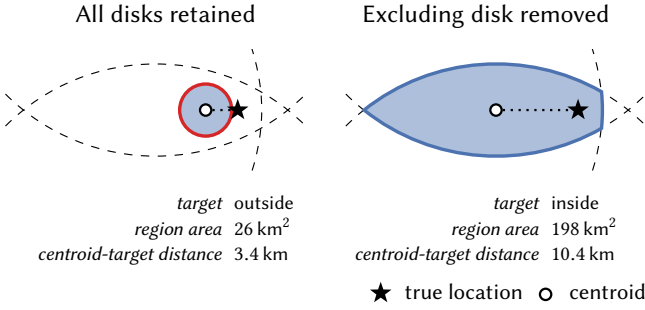


Figure 5: The small red disk defines a small region that only just excludes the target; removing it produces a larger region that does include the target.

0.10% for landmarks between 1000 and 3000 km from the target—a 30-fold difference. We call this the *precision paradox*.

We speculate this paradox arises because “last hop” delays during calibration can cause a landmark’s bestline disks to be too small by a constant distance. Because it’s constant, this error is most significant when the disk itself is small. The median shortfall is 7 km for disks generated by landmarks within 50 km of the target, and 44 km for landmarks farther than 3000 km from the target, but this corresponds to a *relative* shortfall of 82% for nearby landmarks and 1% for distant landmarks.

Since both the algorithms we tried were failures, we decided to test original CBG++ equipped with an oracle that removes exactly the disks that exclude the target. We could not deploy this oracle algorithm against targets in unknown locations, but it lets us estimate the side effects of filtering out bad disks, and infer whether hunting for a workable algorithm would be worthwhile. 100% in-region containment comes only at significant cost: the predicted regions are enlarged for 93.4% of targets.

Predicted regions enlarge when bad disks are removed because incorrect disks are often close enough to the true location to improve the quality of CBG++’s predictions according to the precision metrics, at the cost of the accuracy ones. Incorrect disks generated by landmarks within 50 km of the target have a median radius of only 2.6 km, and the true locations of targets that were not contained lie a median of only 11.5 km outside the predicted region. Figure 5 illustrates the phenomenon.

We infer that exclusion of overly small disks is *not* worthwhile; effort would be better spent on avoiding their creation. If “last hop” delays could be estimated for each landmark, perhaps using route traces as in Octant [21], that might help. Intersection algorithms that explicitly allow for uncertainty should also help, by mitigating the effect of a small disk.

6 Related Work

Several systems have extended CBG by changing what information is combined or how it is represented. TBG [14] adds network topology, using intermediate routers as additional constraints. Octant [21] adds both positive and negative constraints and supports convex and concave regions rather than only disks. Spotter [15] takes a different approach: it models the RTT–distance relationship probabilistically and combines the likelihoods from multiple landmarks. Our results give another reason to consider this approach:

even highly reliable individual constraints can fail when many of them are combined.

More recent work has emphasized finding a landmark close to the target rather than combining many landmarks [6, 8, 12, 13], which helps only where such a landmark exists (Section 1).

Active geolocation has also been studied in adversarial settings. Muir and van Oorschot [17] consider both evasion and counterevasion. Gill et al. [10] show how measurement-based geolocation can be evaded, and Abdou et al. [1] show how manipulating network delay can control the location inferred by delay-based geolocation.

7 Ethics

This study was conducted using only publicly available data from the RIPE Atlas database. We did not conduct any additional active measurements. We only geolocated hosts (the Atlas anchors) dedicated to the purpose of Internet measurement research, whose true physical location is public information.

Active geolocation can be used to reveal locations of hosts whose operators did not wish for that information to be revealed. This may be either ethical or unethical, depending on who the operators are and what the host is being used for. Revealing the location of a botnet’s command-and-control server may be an ethical and valuable act of investigative journalism, but revealing the minute-by-minute location of an individual’s mobile phone is an unethical breach of privacy. Some researchers have considered how to *prevent* hosts from being actively geolocated [1, 10, 17]. Prior work shows that this is possible, but may require defenses beyond simply routing traffic through a VPN or Tor.

8 Conclusion

We began this study with the hypothesis that tighter per-landmark constraints would improve localization. We found just the opposite: tighter constraints cause per-landmark distance estimates to be too short, and CBG++ recovers by falling back to much larger *bestline* estimates, producing poorer localizations overall. The precision paradox suggests, in fact, that calibration should be *loosened* at short ranges to account better for distance-independent delays.

We then turned to the possibility of improving on CBG++’s rejection of disks that are too small, since nearly a quarter of final regions fail to contain the target, and a tiny fraction of disks (0.28%) are responsible for all those failures. We found that these erroneous disks still provide useful information, because they are only a little bit too small. Removing them—even with perfect knowledge of which ones they are—improves accuracy but harms precision. This suggests that future work should concentrate on disk combination techniques that allow for a middle ground between treating each constraint as accepted or discarded.

We used RIPE Atlas anchors as both landmarks and evaluation targets. Anchors are institutionally hosted, well-connected, and concentrated in Europe and North America; our findings may not generalize to residential or mobile targets, or to other continents. Our findings also have not been tested against the scenario that motivates active geolocation in the first place: accurate location of hosts whose operators are trying to manipulate the locations reported by passive geolocation.

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